



32-Hour Workweek Act

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32-Hour Work Week Act

- Would require employees to pay overtime pay (at least 1.5 times the regular wage) for hours worked beyond 32 hours per week.
- Phased in over three years
 - Year 1: 40 to 36
 - Year 2: 36 to 34
 - Year 3: 34 to 32



Senator Barrasso R-WY

- Known to vote in favor of republican based-bills
- Known to vote in favor of blue collar
- Known to vote in favor of farmers and ranchers
 - "Grazing is an integral part of keeping Wyoming's rangelands healthy. Ranchers need more flexibility to adapt and respond to on-the-ground threats like drought and wildfires. This will both support our ranching communities and encourage better management of our federal lands. Our bill gives ranchers the tools they need to best manage the grazing activities of their livestock."
- Known to vote in favor of the coal mining industry
 - *"It's been one roadblock after another for the last eight years. Layer after layer of red tape, strangling the coal industry and coal miners."*



Agricultural Industry

- One of the largest industries in Wyoming
- 3% of the state's GDP & supports large portion of Wyoming's labor force
- Capital & Labor are more substitutable (Cobb-Douglas)
- On average work 50-60 hours per week





Coal Mining Industry

- Largest industry in Wyoming
- 14% of the state's GDP and 40% of the total coal used in the US, large portion of Wyoming's labor force
- Fixed Capital & Labor (Perfect Complements)
- On average work 50-60 hours per week

Coal Mining

- Production Function: $q = \min \{\alpha K, \beta L\}$, where the parameters α and β represent the fixed proportions in which capital (K) and labor (L) must be used to produce output q .
- Initially, the optimal input demands are $K^* = \frac{q}{\alpha}$ and $L^* = \frac{q}{\beta}$ and the total cost function, $TC = wL^* + vK^* = q \left(\frac{w}{\beta} + \frac{v}{\alpha} \right)$.
- With the policy change, overtime payment (1.5 times the normal wage) begins after 32 hours instead of 40. For a 40-hour workweek, workers earn $\$w$ per hour for the first 32 hours, totaling $32w$. For the remaining 8 hours, they earn $\$1.5w$, totaling $12w$. The total wage for the week becomes $32w + 12w = 44w$ and the new effective hourly wage becomes $w' = \frac{44}{40} w = 1.1w$. So, the new effective hourly wage is 10% higher than the standard wage.
- Since workers are now assumed to produce the same output in 32 hours as they did in 40 hours, their productivity increases by a factor of $z = \frac{\text{original hours to produce } q}{\text{reduced hours to produce } q} = \frac{40}{32} = 1.25$ or 25%. With labor becoming more productive, capital becomes the limiting factor, as there is no substitutability between inputs.

Coal Mining

- While the increase in the effective wage will be reflected in the new total cost TC' , the productivity change will not affect the production function, as the isoquant is non-differentiable. More productive labor cannot produce more than the original output with the same capital due to their fixed ratio. Hence, the input demand functions remain the same with the same output, $K^* = \frac{q}{\alpha}$ and $L^* = \frac{q}{\beta}$, while the new total cost function becomes,
$$TC' = w'L^* + vK^* = 1.1wL^* + vK^* = q \left(\frac{1.1w}{\beta} + \frac{v}{\alpha} \right).$$
 For any values $\alpha, \beta, w, v, q > 0$ here, $TC' > TC$.
- With this new bill, the total cost for miners would increase without any change in their output despite increased productivity of labors. Based on this, we would advise the senator to vote against the bill, considering the potential negative impact on the miners' well-being.

Agriculture

- Production Function: $q = K^a L^b$, where the parameters $0 < a < 1$ and $0 < b < 1$ represent the output elasticities of capital and labor, respectively. These elasticities measure the percentage change in output (q) resulting from a 1% change in the respective input (capital or labor), holding the other input constant. Keeping a and b within 0 to 1 range reflects diminishing marginal returns, which is typical in real-world production processes where increasing inputs leads to smaller incremental gains in output. We also assume $a + b = 1$ or constant returns to scale for this production process which is a reasonable assumption for many industries, suggesting that input proportions can be scaled without loss of efficiency, particularly in competitive markets. So, if the farmers double both capital and labor, output will also double.
- Setting the Lagrangian, $\mathcal{L} = wL + vK + \lambda (q - K^a L^b)$. Deriving the first order conditions and solving for the optimal

inputs, $K^* = \left(\frac{aw}{bv}\right)^{\frac{b}{a+b}} q^{\frac{1}{a+b}} = \left(\frac{aw}{bv}\right)^b q$ and $L^* = \left(\frac{bv}{aw}\right)^{\frac{a}{a+b}} q^{\frac{1}{a+b}} = \left(\frac{bv}{aw}\right)^a q$ and total cost,

$$TC = wL^* + vK^* = \frac{1}{a} \left(\frac{b}{a}\right)^{-b} w^b v^a q.$$

Agriculture

- With the new cost (due to the bill) $w' = 1.1w$ and the increased labor productivity $z = 1.25$, the labor demand will change and since there is substitutability between capital and labor, the capital demand will also potentially change to produce the same output, q as before and the new capital would be K' , labor would be L' and incorporating the productivity factor, z , the new production function would be $q = K'^a(zL')^b = 1.25^b K'^a L'^b$ and the new Lagrangian will be, $\mathcal{L}' = 1.1wL' + vK' + \lambda (q - 1.25^b K'^a L'^b)$.

- Solving for the input demand functions,

$$K'^* = 1.1 (1.1)^{-a} (1.25)^{-b} \left(\frac{aw}{bv} \right)^b q = 1.1 (1.1)^{-a} (1.25)^{-b} K^*$$

$$\text{and similarly, } L'^* = (1.1)^{-a} (1.25^{-b}) \left(\frac{bv}{aw} \right)^a q = (1.1)^{-a} (1.25^{-b}) L^*.$$

- Here, $K'^* < K^*$ and $L'^* < L^*$ for any permissible values of a and b (between 0 and 1).

$$\text{For } a = b = 0.5: K'^* = 0.94 K^*, L'^* = 0.85 L^*$$

$$\text{For } a (=0.3) < b (=0.7): K'^* = 0.91 K^*, L'^* = 0.83 L^*$$

$$\text{For } a (=0.7) > b (=0.3): K'^* = 0.96 K^*, L'^* = 0.87 L^*$$

Agriculture

- This relative sequence of results holds for every value of $0 < a < 1$ and $0 < b < 1$. We can see that as labor becomes more productive and expensive, the demand for both inputs falls to produce the same output q , while the demand for labor decreases by a slightly larger factor than the demand for capital due to the direct impact of wage increases.
- With L^* and K^* , the new total cost,

$$TC' = \left(\frac{1}{a} + 0.1\right)(1.1)^{-a}(1.25)^{-b} \left(\frac{b}{a}\right)^{-b} w^b v^a q = (1+0.1a) (1.1)^{-a}(1.25)^{-b} TC.$$

Where $TC' < TC$ for any permissible values of a and b .

For $a = b = 0.5$: $TC' = 0.90 TC$.

For $a(=0.3) < b(=0.7)$: $TC' = 0.86 TC$.

For $a(=0.7) > b(=0.3)$: $TC' = 0.94 TC$.

- Thus, the total cost for farmers decreases at the optimal level without changing the production amount under the policy. Therefore, it is advisable to vote in favor of the bill to improve farmers' well-being.



Final Decision

- Senator Barrasso should vote NO on the bill
- Impacts coal industry positively
- Impacts agricultural industry more negatively
- But coal industry has a higher impact on Wyoming